

MedAssist AI

Frontend & Data Visualization Milestone 1 Report

Prepared By: AKANKSHA MUMMANA

B.Tech – Computer Science & Engineering

Role: Frontend Developer & Data Visualizer

Internship: Infosys Springboard Internship

Project: MedAssist AI – AI Powered Healthcare Prediction System

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1. Project Description

MedAssist AI is an AI-powered healthcare web application designed to assist users in identifying possible diseases based on the symptoms they provide. The application offers a simple, responsive, and user-friendly interface that enables users to enter their symptoms, view AI-generated disease predictions, access previous reports, and manage their personal health profile.

For this milestone, my assigned role was Frontend Developer & Data Visualizer. My primary responsibility was to design and develop the patient dashboard interface and landing portal layouts, ensuring that the application is responsive, modular, and visually interactive. I focused on integrating Chart.js graphics to display disease analytics and configuring timezone formatting so the frontend aligns dynamically with the user's local timezone.

This milestone successfully delivered the core user interface layouts, interactive visualization modules, and client-side routing. The codebase was constructed with reusable React components, laying a clean foundation for backend API integration and clinical prediction workflows.

Objectives of this Milestone

- Develop a responsive patient dashboard interface using React.js and Vite.
- Create reusable UI components like Navigation bars, cards, and layouts.
- Integrate Chart.js to build visual data representation for disease statistics.
- Implement UTC to local Time Zone (IST) formatting for all report timestamps.
- Design responsive Landing Page, Login, and Registration interfaces.
- Set up client-side routing and navigation using React Router DOM.

2. Technologies Used

The MedAssist AI frontend was developed using modern web technologies to ensure scalability, responsiveness, and maintainability.

Technology	Purpose
React.js	Building reusable frontend components and routing layouts
Vite	Fast development server and optimized build tool
React Router DOM	Client-side navigation between modules
Chart.js / react-chartjs-2	Rendering interactive Pie and Line charts for health trends
Tailwind CSS	Styling responsive UI layouts and cards
Local Storage	Persisting user login sessions and dashboard states
Git & GitHub	Version control and collaboration
Visual Studio Code	Source code editor

3. Environment Setup

Development Environment: Visual Studio Code, Node.js & npm, React + Vite, Git & GitHub

Commands used:

```
npm install  
npm run dev
```

4. Frontend Architecture

Implemented modular folder structure: components/ pages/ services/ styles/ layout/

A reusable Layout component is implemented with Sidebar Navigation and a shared UI frame to ensure consistency across the application.

5. Landing Page Development

Implemented a fully responsive landing page including Hero, About, Features, How It Works, Testimonials, and Footer sections. The design is styled with Tailwind CSS utility classes and includes a modern healthcare illustration integrated into the layout.

6. Authentication Module

Designed secure, responsive Login and Registration pages. The structure is configured to connect with JWT-based backend endpoints, storing user session keys upon authentication.

7. Dashboard Module

Developed the Patient Dashboard with: Sidebar Navigation, Statistic Cards, Recent Activity Panel, and disease analytics using **Chart.js Graphical Visualization**:

- Pie Chart Integration : Added a Pie Chart to display the distribution of predicted diseases, allowing users to view the proportion of different conditions in their diagnosis history.
- Line Chart Integration : Added a Line Chart tracking symptom occurrences and frequency trends over time (monthly mapping) to provide insights into patient health trends.

8. Symptom Checker Module

Created the initial symptom selection form interface. The page allows users to enter patient metrics: Age, Gender, Height, Weight, and select multiple symptom indicators flatly from the list before navigating to the prediction result page.

9. AI Prediction Module

Displays selected symptoms, predicted disease, confidence score, severity level, and AI recommendations. Currently connected using frontend mock data schemas.

10. Reports Module

Created the Reports page layout for displaying previous patient assessment logs. Time zone formatting logic was implemented inside the table rows, converting standard ISO UTC date strings to the user's local timezone (IST) for accurate record logs.

11. Profile Management

Implemented editable profile page allowing patients to update their details and upload a profile picture, with state persisting across page refreshes using browser Local Storage.

12. Challenges & Solutions

Challenges Faced

- Integrating Chart.js canvas containers inside responsive flex grids caused scaling issues on smaller viewport sizes.
- Datetime strings returned from mock endpoints default to UTC, causing timestamps in the reports table to mismatch the user's actual local clock.

Solutions Implemented

- Configured Chart.js responsive options, added CSS wrapper containers, and enabled `maintainAspectRatio` to `false`.
- Used JavaScript's `Intl.DateTimeFormat` and `Date` object methods to convert UTC datetime strings into local timezone strings (IST) dynamically.

13. Conclusion

The Frontend Development and Data Visualization milestone for the MedAssist AI project has been completed successfully. A responsive React-based interface was developed, integrating interactive Pie and Line charts for graphical health tracking. Client-side timezone conversions were implemented to align report history logs with the user's actual local time. This milestone establishes a clean, visually informative structure ready for backend API integration and live machine learning model deployments.

Outcome

- Successfully developed a responsive React-based frontend using Vite.
- Implemented Chart.js Pie Chart for disease distribution analytics.
- Implemented Chart.js Line Chart for monthly symptom frequency trends.

- Configured local time zone date-time formatting for reports history.
- Created client-side routing using React Router DOM.